

```

#include <stdio.h>

struct Process {
    int pid, at, bt;
    int ct, tat, wt, rt;
};

int main() {
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("\nProcess %d\n", p[i].pid);
        printf("Arrival Time: ");
        scanf("%d", &p[i].at);
        printf("Burst Time: ");
        scanf("%d", &p[i].bt);
    }

    int time = 0;
    float avg_tat = 0, avg_wt = 0, avg_rt = 0;

    for(int i = 0; i < n; i++) {

        if(time < p[i].at)
            time = p[i].at;

        p[i].rt = time - p[i].at;
        time += p[i].bt;

        p[i].ct = time;
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt = p[i].tat - p[i].bt;

        avg_tat += p[i].tat;
        avg_wt += p[i].wt;
        avg_rt += p[i].rt;
    }

    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\tRT\n");

```

```
for(int i = 0; i < n; i++) {  
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",  
        p[i].pid, p[i].at, p[i].bt,  
        p[i].ct, p[i].tat, p[i].wt, p[i].rt);  
}  
  
printf("\nAverage Turnaround Time = %.2f", avg_tat/n);  
printf("\nAverage Waiting Time = %.2f", avg_wt/n);  
printf("\nAverage Response Time = %.2f", avg_rt/n);  
  
return 0;  
}
```

```
C:\Users\Admin\Desktop\1  X + v
Enter number of processes: 3

Process 1
Arrival Time: 2
Burst Time: 5

Process 2
Arrival Time: 3
Burst Time: 2

Process 3
Arrival Time: 4
Burst Time: 3

PID    AT    BT    CT    TAT    WT    RT
1       2     5     7     5     0     0
2       3     2     9     6     4     4
3       4     3    12     8     5     5

Average Turnaround Time = 6.33
Average Waiting Time = 3.00
Average Response Time = 3.00
Process returned 0 (0x0)  execution time : 21.476 s
Press any key to continue.
|
```

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